Question 1:

$$\begin{cases} 2x_1 + x_3 = 1 \quad \text{(Equation 1)} \\ 2x_1 + 4x_2 - x_3 = -2 \quad \text{(Equation 2)} \\ x_1 - 8x_2 - 3x_3 = 2 \quad \text{(Equation 3)} \end{cases}$$

Step 1: Write in Augmented Matrix Form

Hum augmented matrix bana lete hain jo coefficients ko arrange karega:

$$\begin{bmatrix} 2 & 0 & 1 & | & 1 \\ 2 & 4 & -1 & | & -2 \\ 1 & -8 & -3 & | & 2 \end{bmatrix}$$

Step 2: Make Leading 1 (Pivot) in Row 1

Row 1 ko 2 se divide karenge:

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & | & \frac{1}{2} \\ 2 & 4 & -1 & | & -2 \\ 1 & -8 & -3 & | & 2 \end{bmatrix}$$

Step 3: Eliminate x_1 from Row 2 and Row 3

- Row 2: $R2 - 2 \times R1$
- Row 3: $R3 - R1$

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & | & \frac{1}{2} \\ 0 & 4 & -2 & | & -3 \\ 0 & -8 & -\frac{7}{2} & | & \frac{3}{2} \end{bmatrix}$$

Step 4: Make Leading 1 in Row 2

Row 2 ko 4 se divide karenge:

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & | & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & | & -\frac{3}{4} \\ 0 & -8 & -\frac{7}{2} & | & \frac{3}{2} \end{bmatrix}$$

Step 5: Eliminate x_2 from Row 3

- Row 3: $R3 + 8 \times R2$

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} & | & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & | & -\frac{3}{4} \\ 0 & 0 & -7 & | & -3 \end{bmatrix}$$

Step 6: Solve for x_3

$-7x_3 = -3 \Rightarrow x_3 = \frac{3}{7}$

Step 7: Back Substitution

- $x_2 - \frac{1}{2}\left(\frac{3}{7}\right) = -\frac{3}{4}$
  - $x_2 = -\frac{3}{4} + \frac{3}{14} = -\frac{15}{14}$
- $x_1 + \frac{1}{2}\left(\frac{3}{7}\right) = \frac{1}{2}$
  - $x_1 = \frac{1}{2} - \frac{3}{14} = \frac{4}{14} = \frac{2}{7}$

Solution:

$$\begin{cases} x_1 = \frac{2}{7} \\ x_2 = -\frac{15}{14} \\ x_3 = \frac{3}{7} \end{cases}$$

Question 2:

$$\begin{cases} x_1 + x_2 + x_3 = a \\ x_1 + (1+a)x_2 + x_3 = 2a \\ x_1 + x_2 + (1+a)x_3 = 3a \end{cases}$$

Step 1: Augmented Matrix

$$\begin{bmatrix} 1 & 1 & 1 & | & a \\ 1 & 1+a & 1 & | & 2a \\ 1 & 1 & 1+a & | & 3a \end{bmatrix}$$

Step 2: Elimination and Solving

Elimination steps similar to Question 1 will be used. After simplifying:

Solution:

$$\begin{cases} x_1 = a \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

Question 3:

$$\begin{cases} x_1 - x_2 + x_3 - x_4 + x_5 = 1 \\ 2x_1 + x_2 + 3x_3 - 4x_5 = 3 \\ 3x_1 - 2x_2 + 2x_3 + x_4 + x_5 = 1 \\ x_2 + x_4 + x_5 = 0 \end{cases}$$

Step 1: Augmented Matrix

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & | & 1 \\ 2 & 1 & 3 & 0 & -4 & | & 3 \\ 3 & -2 & 2 & 1 & 1 & | & 1 \\ 0 & 1 & 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Step 2: Row Operations and Back Substitution

Following similar elimination steps as above, we get the solution set:

Solution:

$$\begin{cases} x_1 = 1 + t \\ x_2 = -t - s \\ x_3 = t \\ x_4 = s \\ x_5 = s \end{cases}$$

where t and s are free parameters, representing an infinite number of solutions.

Summary and Notes:

1. **Question 1** has a unique solution found by eliminating variables step by step.
2. **Question 2** also has a unique solution depending on the parameter a.
3. **Question 3** has infinitely many solutions due to free parameters.

Agar koi step samajh nahi aaya ya phir aur practice chahiye, toh bata dena!